

Learning goal 1:

Can navigate the terminal: `ls`, `cd`, `whoami`, `pwd`.

I can do it!

`ls` = list (show a list of the folder your in)

`cd` = change directory (for example if have a directory on your desktop named documents you can say `cd documents` and open the folder)

`whoami` = who am i (Shows the user logged in to the terminal/pc)

`pwd` = print working directory (Shows you which directory you are currently in)

Learning goal 2:

Can create, delete files / folders: `mkdir`, `rm`, `cp`, `mv`, `touch`, `nano`.

I can do it!

`mkdir` = Create / make directory (for example `mkdir hej` now you have a folder name hej)

`rm` = delete file/directory (For example `rm hej` now you no longer have a folder called hej)

`cp` = copy (self explanatory)

`mv` = move (for example `mv files` to a different directory)

`touch` = create file (`touch hej` now you have a new file called hej)

`nano` = edit file (`nano hej` now you can edit the contents of hej)`cd`

Learning goal 3

Can use the following basic terminal commands: `cat`, `wc`, `uniq`, `sort`

I can do it!

`cat` = Shows the contents of a file

`wc` = word count counts words / lines

`uniq` = unique, removes duplicates

`sort` = sort, also self explanatory

Learning goal 4

Can create a new repository in your preferred git provider:

I can do it!

Learning goal 5

Can perform basic git operations: clone, add, commit, push, pull

I can do it!

git clone x

git add -A or git add (Specific file)

git commit -m "commit message"

git push (all or you can chose specifics)

git pull

Learning goal 6

Can list the main hardware components of a computer, their functions and the von neumann architecture

Main components:

CPU = data processer

RAM = temporary memory

Storage = Permanent memory

Input/Output = user interaction

Von Neumann architecture:

there is a input device (fx keyboard) which sends a signal to the CPU where the control unit and logic unit is, it communicates with the memory unit and sends it to the output device (Viewing screen)

Learning goal 7

Can explain how computers work, starting from hardware all the way to software

Hardware executes operations

Operating system manages resources

software provides functionality

Learning goal 8

Can talk about processes in operating systems

a process is a running program

Each process has its own memory

the operating system manages CPU time

Multiple processes can run at the same time

Learning goal 9

Can talk about different number representations such as: Binary, hexadecimal, decimal

I can do it!

Binary = base 2

Decimal = base 10

Hexadecimal = base 16

Learning goal 10

Can bring up real-world use cases for different number representations

Binary is used by computers

Hexadecimal is used in debugging and memory addresses

Decimal is numbers that readable for humans

Learning goal 11

Can explain different charsets like ASCII and Unicode and how they differ

ASCII

128 characters

Limited

Unicode

Supports many languages

modern standard

Learning goal 12

Knows about different types of databases and their use cases

SQL is for structured data

NoSQL is for flexible data

Learning goal 13

Can setup a new MySQL database and connect to it

I can do it!

Create database
connect using JDBC

Learning goal 14
Can create DDL statements to create tables

I can do it!

```
Create table users (  
id INT PRIMARY KEY,  
name VARCHAR(20)  
);
```

Learning goal 15
Can create DML (SELECT, INSERT) statements.

```
SELECT * FROM users;
```

```
INSERT INTO users VALUES (1);
```

Learning goal 16
Can use WHERE clause to filter results.

I can do it!

```
SELECT * From users WHERE id = 1;
```

Learning goal 17
Can define Primary Keys

a primary key uniquely identifies a row and cannot be duplicated

Learning goal 18
Understands how Foreign Keys work

foreign keys links different tables together

Learning goal 19
In SELECT statements, can use (with moderate help) LIMIT, ORDER BY, GROUP BY
Aggregate functions like COUNT, SUM, AVG, MIN and MAX
Pattern matching with LIKE and wildcards
JOIN to combine data from multiple tables

```
SELECT * FROM users LIMIT 5;  
SELECT * FROM users ORDER BY username ASC;  
SELECT name, COUNT(*) AS antal FROM users GROUP BY name;
```

```
SELECT COUNT(*) AS antal FROM users;
SELECT AVG(age) AS average_age FROM users;
SELECT SUM(price) AS total FROM orders;
SELECT MIN(price), MAX(price) FROM products;
SELECT * FROM users WHERE username LIKE 'a%'; (Usernames starting with a)
SELECT * FROM users WHERE username LIKE '%sen'; (Usernames ending in sen)
```

Inner join:

```
SELECT users.username, orders_id FROM users INNER JOIN orders ON users.id =
orders.user_id;
```

Left join:

```
SELECT users.username, orders.id FROM users LEFT JOIN orders ON users.id =
orders.user_id;
```

Learning goal 20

Can create inner, left and right joins after looking up the syntax

I can do it!

Inner join:

```
SELECT users.username, orders_id FROM users INNER JOIN orders ON users.id =
orders.user_id;
```

Left join:

```
SELECT users.username, orders.id FROM users LEFT JOIN orders ON users.id =
orders.user_id;
```

Right join:

```
SELECT users.username, orders.id FROM users RIGHT JOIN orders ON users.id =
orders.user_id;
```

Learning goal 21

Can create DDL statements to create tables with constraints: PRIMARY KEY,
AUTO_INCREMENT, FOREIGN KEY, UNIQUE, NOT NULL

```
CREATE TABLE users (
id INT AUTO_INCREMENT PRIMARY KEY, username VARCHAR(255) UNIQUE NOT
NULL, password VARCHAR(255) NOT NULL );
```

Learning 22

Can create DML (UPDATE, DELETE) statements

I can do it!

```
UPDATE users SET username = 'newname' WHERE id = 1;
```

```
DELETE FROM users WHERE id = 1;
```

Note: be very careful with syntax as just writing DELETE FROM users; will delete all rows

Learning goal 23

Can create a pull request

Steps 1 create a branch on your project

step 2 push changes to the project

step 3 open pull request on GitHub

step 4 review the push and merge

Learning goal 24

Understands different Git workflows such as GitHub Flow

GitHub Flow is a simple workflow where you create a branch, make changes, create a pull request, and then merge into main.

Learning goal 25

Can solve a merge conflict

I can do it!

different text on same line

choose what to keep as in manually edit the file and commit the resolved version

Learning goal 26

Can write YAML

name: Build project

on: push

jobs:

build:

runs-on: ubuntu-latest

steps:

- run: mvn clean install

Note: spacing is very important in YAML

Learning goal 27

Understands what GitHub actions are and can breakdown workflows into runners, jobs and steps

Runner is the machine like in the example above its ubuntu

Jobs is the tasks like build, test or deploy as we used in ønskeskyen

Steps is like things to do within jobs

Learning goal 28

Can give use cases for GitHub Actions

There are many use cases for github actions like build project, run tests, deploy to azure or automate workflows

Learning goal 29

Understands different cloud service models: IaaS, PaaS, SaaS

Learning goal 30

Can deploy a web application to Azure app service

I can do it!

you push code to your github repository and if you have a CD (continuous deployment) it will automatically send the newest version of the main branch to azure to update your website

Learning goal 31

Understands the flow from pushing, Github actions running, building the project and deployment to Azure

github push -> github actions -> build -> deploy -> azure runs app

Learning goal 32

Adding a database to a Spring Project

use JDBC to create connection - use repositories - connect to the database

and you can use environmental variables to make it even safer (or github secrets depending on the project)

Learning goal 33

Setting up a database in Azure

create the database on azure

make a username (cant be root) and a password

and allow access to certain ips you want to have access to the database

Learning goal 34

Can set up a database in Azure and connect it to azure app service project with guide

I can do it!

Create a database in Azure, set environment variables (DB_URL, DB_USER, DB_PASSWORD), and connect using JDBC.

Learning goal 35

Is able to go through scenarios that can cause concurrency problems in databases

problems typically happen when multiple people update data at the same time this can result in lost or incorrect data in the database

Learning goal 36

Can explain what ACID is and why it solves concurrency problems for databases

Atomicity - all or nothing (no half processes)

Consistency - makes sure data is always valid

Isolation - separate everything

Durability - data is saved forever

as long as you follow those it should prevent errors in your databases

Learning goal 37

is aware of the possibility to define transactions in SQL and JDBC

I can do it!

Transactions can be used in SQL and JDBC to ensure safe operations.

A transaction is when multiple database operations are grouped together, and they either all succeed or all fail.

Example:

```
BEGIN;
```

```
UPDATE accounts SET balance = balance - 100 WHERE id = 1;
```

```
UPDATE accounts SET balance = balance + 100 WHERE id = 2;
```

```
COMMIT;
```

ROLLBACK can be used if something goes wrong.

Atomic means all or nothing. If one part fails, everything is rolled back.